Irbaaz Ahmed Patel

Robotics · Perception · Manipulation

Email · +44 7553684548 · GitHub · LinkedIn

Eligible to work in the UK without sponsorship (Graduate Route) · Available immediately

Robotics MSc with a background in AI and machine learning. My dissertation put a language-instructed grasping pipeline on a real Toyota HSR, so most of my time went on the unglamorous half of robotics: calibration, reachability, and why a plan that runs in simulation fails on hardware. After a perception or manipulation role where the robot is physical.

TECHNICAL SKILLS

Languages	Python, C++, Embedded C, SQL, MATLAB · Linux, Git, Docker, CMake, CUDA
Robotics & ML	ROS 2 Jazzy, ROS 1 Noetic, Gazebo, PyBullet, CARLA · PyTorch, OpenCV, TensorRT, ONNX, deep RL

EDUCATION

MSc Robotics · Heriot-Watt University, Edinburgh *2025 – 2026*

B.Tech in Artificial Intelligence and Machine Learning · NMIMS *2020 – 2025*

Certifications: Modern Robotics (Northwestern) · Machine Learning with Python, Deep Learning with Keras, GenAI Applications (IBM).

PROJECTS

Instruction-Conditioned Robotic Grasping on a Toyota HSR

MSc Dissertation, National Robotarium

ROS Noetic · GPT-4o · Grounding DINO + SAM · Contact-GraspNet · camera calibration · visual servoing

- Built an end-to-end pipeline turning a plain-language instruction into a grasp on real hardware: a vision-language model reasons about which part to grasp, open-vocabulary segmentation grounds it in the image, Contact-GraspNet proposes poses.
- Ran **11 hardware trials** including four consecutive autonomous grasps; object identification **94% (61/65)**.
- Diagnosed the reach failures blocking early runs: collision-aware base placement took reachable grasps **from 0 of 25 to 25 of 25**, and visual servoing converged from 46.9 px to 12.7 px.
- Main finding: language names part-level constraints segmentation cannot resolve (masks collapse to 86–97% whole-object coverage); point-cloud decomposition recovers that structure.
- Wrote the lab safety protocol the project now runs under, checking every arm pose against the URDF for joint limits, floor clearance and a head keep-out zone. Supervised by Dr. Mauro Dragone.

Real-Time Object Detection Node in ROS 2

C++17 · ROS 2 Jazzy · rclcpp · ONNX Runtime · cv_bridge

Ported my Python perception work to C++: an ament_cmake package whose class-based rclcpp node takes a camera stream through cv_bridge, runs YOLOv8 inference via the ONNX Runtime C++ API, and publishes detections to RViz2, with per-frame latency measured.

Deep RL for Robotic Pick-and-Place

Stable-Baselines3 · TQC + HER · panda-gym · PyBullet

Trained a Truncated Quantile Critics agent with Hindsight Experience Replay on PandaPickAndPlace-v3 to 100% success over 20 evaluation episodes at 9.4 steps each, across 1M timesteps. PPO stalled below 5% on the same sparse reward; HER made the task tractable.

Humanoid Locomotion with Deep RL

Gymnasium · Stable-Baselines3 · reward shaping

Carried the same algorithm across to high-dimensional locomotion, reaching a peak reward of 5,742 through tuning and reward shaping.

LLM-Driven Teaching Assistant on Marty V2

LLM prompting · TTS · behaviour design · study design

Dual-mode behaviour system for a humanoid classroom robot: narrative storytelling and LLM Q&A, with speech, LED and servo coordinated per beat. Designed a three-condition between-subjects evaluation protocol.

Autonomous Driving and State Estimation

CARLA · sensor fusion · attention · Kalman filtering · LiDAR

TransFuser-Lite: camera and LiDAR fused through attention for waypoint prediction, closed-loop in CARLA. EKF localisation fusing wheel odometry with LiDAR, cutting position error 97.5%.

Beach Litter Detection and Path Optimisation for Cleaning Drones

Published — Proc. ICNGT 2025, Routledge (Taylor & Francis)

Drone-mounted detector at 71.4% mAP50 and 70 FPS after TensorRT and ONNX optimisation, paired with a planner for collection routes.

EXPERIENCE

AI Product Development Intern · Clarion Analytics Pte. Ltd.

Jan – May 2025

PyTorch · OpenCV · transfer learning · generative AI

- Built deep learning pipelines to extract structure from P&ID engineering drawings, with custom feature detection for diagrams that standard OCR could not read.
- Developed a fraud detection system for health insurance claims using computer vision and transfer learning, including the annotation workflows and domain-specific model training behind it.

LEADERSHIP, COMMUNITY & INTERESTS

Head of Department (Creative), then Organising Committee · Atrangi Fine Arts Club, NMIMS *2021 – 2024*

Student Grievance Committee · NMIMS Mukesh Patel School of Technology *2023 – 2024*

Research & Development · Narsee Monjee Model United Nations, Shirpur *2022 – 2023*

Volunteer: Human Appeal Scotland, Heriot-Watt Islamic Society, and Developmental and Ecofriendly Enterprises, Mumbai.

Interests: Poetry, visual art, and technology - where robotics and the arts meet for me.